Two pests in a Bt cotton refuge

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1 Overview

Bt cotton is planted with a refuge of non-Bt cotton, where susceptible pests survive and slow the evolution of resistance to Bt. Two caterpillars, *Spodoptera frugiperda* and *Helicoverpa armigera*, compete in such refuges, and insecticides are sprayed there. Malaquias et al. (2021) built a cellular automaton of a refuge from field data on how well each species survives alone and together, at low and high density. They asked which species comes to dominate under five insecticide and resistance scenarios. The model has no Bt plants in it: every cell is a refuge plant, and resistance means resistance to the insecticide.

The source leaves several mechanisms unstated: the encounter and colonisation probabilities, the role of adult emergence, how resistance works, and where the first insects are placed. Its Table 1 also contradicts its Figure 1. This model is an adaptation that fills these gaps in stated ways. It reproduces some of the source's results and not others, and the vignette shows which.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 The model

A 1000×1000 torus of plants, each empty or holding one of the two species, is updated once a generation from its Moore neighbours:

- an occupied plant loses its larva to natural enemies with probability 0.25; then, in a sprayed generation, to insecticide with probability 0.99, unless its species is resistant; then to competition;
- survival in competition depends on density (high if at least four neighbours are occupied) and on whether the larva meets the other species, which it does with probability equal to the share of its neighbours holding that species;
- an empty plant is colonised by *S. frugiperda* with probability `0.86 n₁R₁ / (n₁R₁ + n₂R₂)`, and failing that by *H. armigera* likewise, where `nᵢ` are the neighbours holding each species and `Rᵢ` their fecundities (400 and 725).

The insecticide is sprayed in every generation in which either species occupies at least a fifth of the field. That decision needs the whole field, so it is a callback stored with the problem.

```
model = build(:bt_cotton_refuge)
```

```
CAModel bt_cotton_refuge
  Malaquias et al. (2021) S. frugiperda and H. armigera in a Bt cotton refuge
  states:
    plant::Int8 = 0 ∈ 0:2  # 0 empty, 1 S. frugiperda, 2 H. armigera
  parameters:
    mortality = 0.25  # natural mortality and dispersal
    kill = 0.99  # insecticide mortality
    spray = false  # this generation is sprayed (set by the callback)
    resistant_Sf = false  # S. frugiperda resists the insecticide
    resistant_Ha = false  # H. armigera resists the insecticide
    escape = 1.0  # probability a resistant larva escapes the spray
    threshold = 0.5  # occupied share of neighbours for high density
    fecundity_Sf = 400.0  # reproductive capacity of S. frugiperda
    fecundity_Ha = 725.06  # reproductive capacity of H. armigera
    emergence = 0.86  # adult emergence, scaling colonisation
    sL_Sf = 0.75  # larval survival
    sL_Ha = 0.8  # larval survival
    sL_Sf_Ha = 0.6  # larval survival
    sL_Ha_Sf = 0.49  # larval survival
    sH_Sf = 0.48  # larval survival
    sH_Ha = 0.22  # larval survival
    sH_Sf_Ha = 0.18  # larval survival
    sH_Ha_Sf = 0.47  # larval survival
  rule: LocalRule(<anonymous function>, Moore(1))
```

The survival probabilities, as the source's Figure 1 gives them (the default) and as its Table 1 does:

```
fig1, tab1 = bt_cotton_survival(:figure1), bt_cotton_survival(:table1)
[(parameter=k, figure_1=fig1[k], table_1=tab1[k]) for k in keys(fig1)]
```

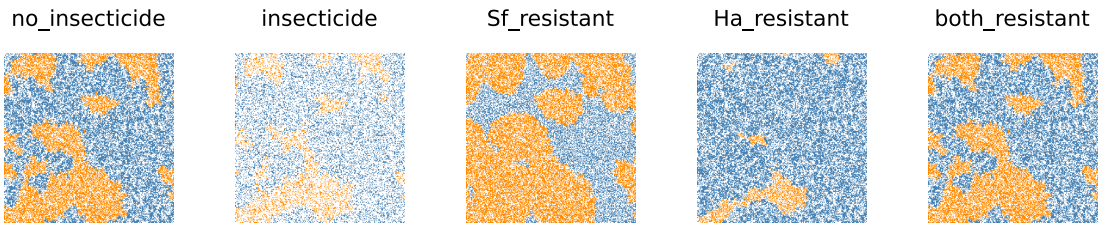
```
8-element Vector{@NamedTuple{parameter::Symbol, figure_1::Float64,
table_1::Float64}}:
 (parameter = :sL_Sf, figure_1 = 0.75, table_1 = 0.63)
 (parameter = :sL_Ha, figure_1 = 0.8, table_1 = 0.46)
 (parameter = :sL_Sf_Ha, figure_1 = 0.6, table_1 = 0.58)
 (parameter = :sL_Ha_Sf, figure_1 = 0.49, table_1 = 0.9)
 (parameter = :sH_Sf, figure_1 = 0.48, table_1 = 0.31)
 (parameter = :sH_Ha, figure_1 = 0.22, table_1 = 0.38)
 (parameter = :sH_Sf_Ha, figure_1 = 0.18, table_1 = 0.11)
 (parameter = :sH_Ha_Sf, figure_1 = 0.47, table_1 = 0.16)
```

At low density *H. armigera* survives a little better alone. At high density *S. frugiperda* does, unless the two meet, when *H. armigera* wins.

3 The five scenarios

Each scenario starts from 500 individuals of each species, scattered at random, and runs for 80 generations:

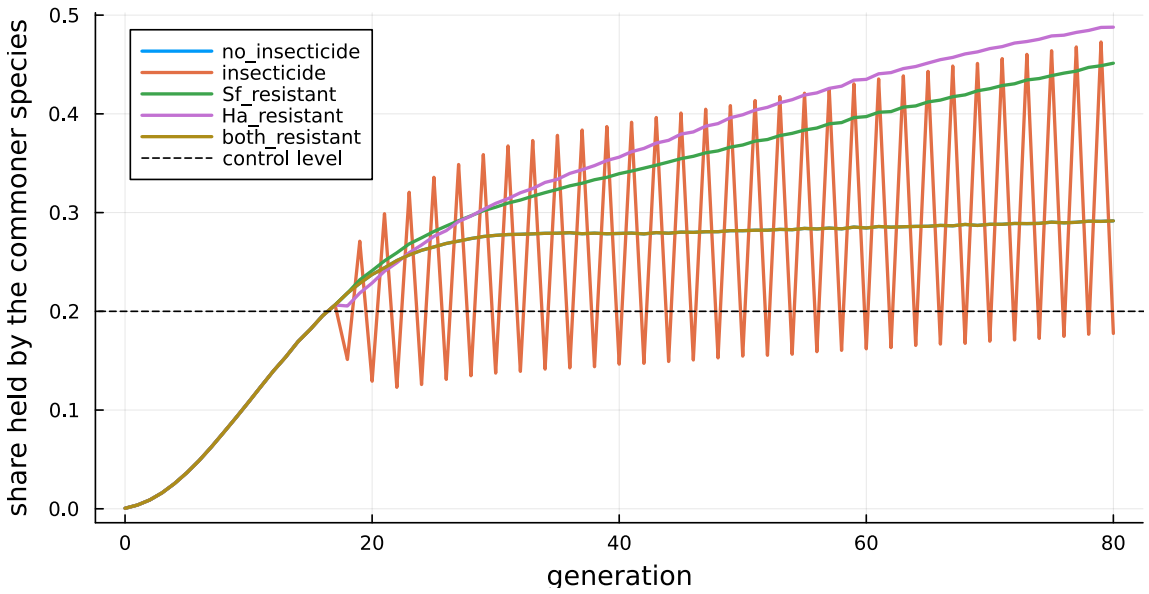
```
scenarios = (:no_insecticide, :insecticide, :Sf_resistant, :Ha_resistant,
:both_resistant)
runs = Dict(sc => solve(problem(:bt_cotton_refuge; scenario=sc, seed=1); saveat=1)
            for sc in scenarios)
fieldmap(u, title) = heatmap(u[1:250, 1:250]; c=[:white, :darkorange, :steelblue],
                             clims=(0, 2), aspect_ratio=:equal, framestyle=:none,
                             colorbar=false, title)
plot([fieldmap(runs[sc].u[end], String(sc)) for sc in scenarios]...; layout=(1, 5),
     size=(1100, 250))
```



A corner of 250 × 250 plants at generation 80: *S. frugiperda* orange, *H. armigera* blue, empty plants white. As in the source's Figure 3, each species holds large connected domains.

Spraying starts when the commoner species reaches the control level, a fifth of the field, around generation 17. Here is the share of the field held by the commoner species:

```
commoner(sol) = [max(count(==(1), u), count(==(2), u)) / length(u) for u in sol.u]
plot(0:80, [commoner(runs[sc]) for sc in scenarios]; lw=2,
     label=permutedims(collect(String.(scenarios))), xlabel="generation",
     ylabel="share held by the commoner species", legend=:topleft, size=(700, 360))
hline!([0.2]; ls=:dash, color=:black, label="control level")
```



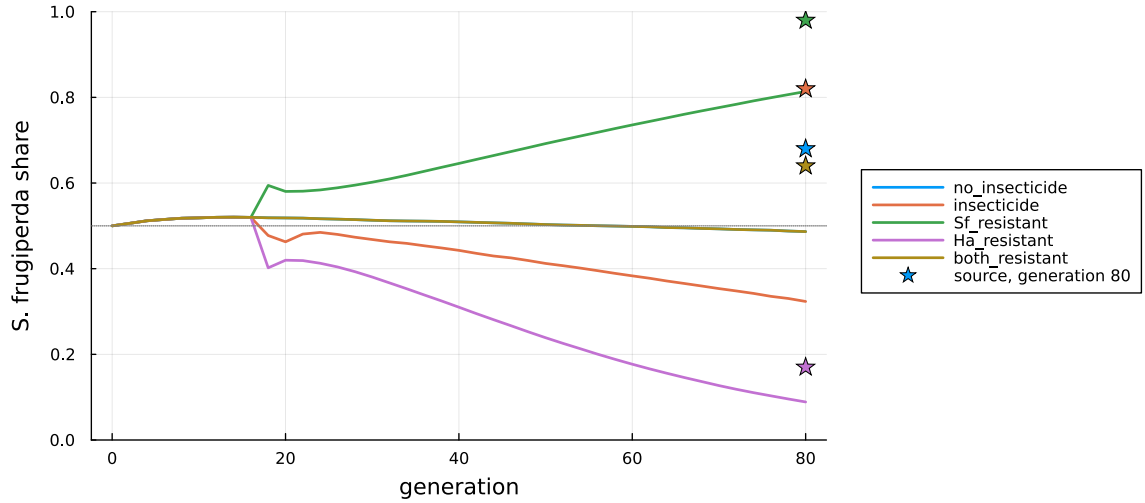
Without resistance, a sprayed generation drops below the control level, the next is not sprayed and climbs back above it, and the field alternates. With one species resistant, the spray removes only the other, and the field fills up again. When both are resistant, the spray changes nothing, and that run is the same as the run without insecticide, whose curve it hides.

4 Dominance

The source's Figure 4 plots each species' share of the occupied plants. Read off that figure, *S. frugiperda* ends at about 0.68, 0.82, 0.98, 0.17 and 0.64 in the five scenarios: it dominates in all but the one where only *H. armigera* is resistant. Three runs each here:

```
dominance(sc; seeds=1:3) = mean([[bt_cotton_dominance(u).Sf
                                  for u in solve(problem(:bt_cotton_refuge;
scenario=sc,
                                                         seed=s); saveat=2).u]
                                 for s in seeds])
dom = Dict(sc => dominance(sc) for sc in scenarios)
source_end = Dict(:no_insecticide => 0.68, :insecticide => 0.82, :Sf_resistant =>
0.98,
                  :Ha_resistant => 0.17, :both_resistant => 0.64)
plot(0:2:80, [dom[sc] for sc in scenarios]; lw=2,
label=permutedims(collect(String.(scenarios))),
```

```
     xlabel="generation", ylabel="S. frugiperda share", legend=:outerright,
ylims=(0, 1),
     size=(850, 380), left_margin=4Plots.mm, bottom_margin=5Plots.mm)
scatter!(fill(80, 5), [source_end[sc] for sc in scenarios]; color=1:5,
label="source, generation 80",
         marker=:star5, ms=7)
hline!([0.5]; ls=:dot, color=:gray, label="")
```




```
[(scenario=sc, model=round(dom[sc][end]; digits=2), source=source_end[sc]) for sc
in scenarios]
```

```
5-element Vector{@NamedTuple{scenario::Symbol, model::Float64, source::Float64}}:
 (scenario = :no_insecticide, model = 0.49, source = 0.68)
 (scenario = :insecticide, model = 0.32, source = 0.82)
 (scenario = :Sf_resistant, model = 0.81, source = 0.98)
 (scenario = :Ha_resistant, model = 0.09, source = 0.17)
 (scenario = :both_resistant, model = 0.49, source = 0.64)
```

The two scenarios in which resistance differs between the species come out as in the source. *S. frugiperda* dominates when only it is resistant, and *H. armigera* when only it is. In the other three scenarios the source has *S. frugiperda* winning, and here *H. armigera* holds slightly more than half without insecticide or with both species resistant, and clearly more with insecticide and no resistance. The first case and the last coincide, as both species escape the spray.

5 The unstated choices

The outcome of competition depends on mechanisms the source does not specify. Four alternative readings, on a 300×300 field without insecticide, three runs each:

```
variants = ["as built" => (;), "Table 1 survival" => bt_cotton_survival(:table1),
            "no emergence factor" => (emergence=1.0,), "equal fecundity" =>
            (fecundity_Ha=400.0,),
            "high density from 5 neighbours" => (threshold=0.625,)]
endshare(pmap) = mean(bt_cotton_dominance(solve(problem(:bt_cotton_refuge; n=300,
initial=45,
                                                        seed=s, pmap);
                                                save_everystep=false).u[end]).Sf
                      for s in 1:3)
[(reading=name, Sf_share_at_80=round(endshare(pm); digits=2)) for (name, pm) in
variants]
```

```
5-element Vector{@NamedTuple{reading::String, Sf_share_at_80::Float64}}:
 (reading = "as built", Sf_share_at_80 = 0.51)
 (reading = "Table 1 survival", Sf_share_at_80 = 0.35)
 (reading = "no emergence factor", Sf_share_at_80 = 0.59)
 (reading = "equal fecundity", Sf_share_at_80 = 0.9)
 (reading = "high density from 5 neighbours", Sf_share_at_80 = 0.49)
```

The colonisation weights decide the outcome. Table 1's survival values favour *H. armigera* still more. With the fecundities the source gives, *H. armigera* colonises empty plants almost twice as readily, and that outweighs its poorer survival at high density. With equal fecundities *S. frugiperda* takes nine tenths of the field. Applying adult emergence to colonisation tilts the balance a little further towards *H. armigera*, and leaving it out gives *S. frugiperda* a slight majority, as in the source's first scenario. The source's results must rest on a colonisation rule that favours *S. frugiperda* more than its description suggests.

6 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:bt_cotton_refuge).deviations)
```

```
- Survival values follow the source's Figure 1 and rule text; its Table 1 gives
different values, available as bt_cotton_survival(:table1).
- The encounter probability is not given in the source; a larva meets the other
species with probability equal to the share of its eight neighbours holding it.
- The colonisation probabilities are not given as formulas; an empty cell is
colonised by S. frugiperda with probability emergence * n1 R1 / (n1 R1 + n2 R2), and
failing that by H. armigera likewise.
- The role of adult emergence (0.86) is not stated in the source; it scales both
colonisation probabilities.
- The resistance mechanism is not specified (the source gives only an initial
resistant frequency of 0.0625 under Hardy-Weinberg equilibrium); a larva of a
resistant species escapes the spray with probability escape (default 1).
- High density means at least half of the eight neighbours occupied (the source
writes both 'greater than or equal to' and 'greater than' 0.5).
- The insecticide acts after natural mortality and before competition; the source
orders it after natural mortality only.
```

```
- The spray decision uses the state at the end of the previous generation.
- The initial individuals are placed on distinct uniformly random cells; the source
does not say where they were placed, nor how many replicates it ran.
```

Malaquias, José Bruno, Daniela R. S. Santana, Paulo E. Degrande, et al. 2021. “Shifts in Ecological Dominance Between Two Lepidopteran Species in Refuge Areas of Bt Cotton.” *Insects* 12 (2): 157. https://doi.org/10.3390/insects12020157.